

Analyzing Non-Normal Data

AGRON 5130

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Introduction

The goal of this assignment is to give you hands-on practice with **diagnosing model assumptions** and working with **non-normal data**.

You will work with a grape yield dataset from a vineyard experiment evaluating canopy management strategies.

In this assignment, you will practice:

- visualizing skewed data
- diagnosing violations of model assumptions
- applying a **log transformation**
- fitting a **GLM**
- interpreting treatment effects across models

Submit a **knitted PDF/HTML generated from your R Markdown file** to Canvas.

Background

Grapevine productivity is strongly influenced by canopy structure, which affects light interception, air movement, and ultimately fruit development.

In commercial vineyards, growers often use different **canopy management practices** to balance vegetative growth and fruit production. However, these practices can have variable effects depending on environmental conditions and vine-to-vine variability.

A field trial was conducted to evaluate the effect of different canopy management strategies on grapevine productivity in a mature vineyard.

The treatments included:

- **Control** (no canopy management)
- **Shoot thinning** (removal of excess shoots early in the season)
- **Leaf removal** (removal of basal leaves around the fruiting zone)
- **Shoot thinning + leaf removal**

The experiment was conducted using a **completely randomized design** with 30 replications.

At the end of the growing season, the number of **grape clusters per vine** was recorded for each experimental unit.

Due to natural variability among vines and the biological processes governing fruit set, the response variable is expected to show:

- substantial variability across treatments
- increasing variance with increasing mean
- right-skewed distributions

These characteristics make this dataset a good candidate for exploring **violations of model assumptions** and methods to address them.

We can start by loading the necessary libraries

```
library(ggplot2)
library(emmeans)
```

Question 1

1 point

Load the dataset "data/grape.csv" and display the first few rows.

note: make sure to call the dataset **grape**. This is important for the models I have pre built for you after question 10.

Question 2

2 points

Create a plot showing the relationship between **treatment** and **clusters**.

Question 3

1 point

Create a histogram of **clusters**.

Briefly describe the distribution (normal vs skewed).

Question 4

2 points

Fit a linear model and extract the treatment means. Then, comment on the confidence intervals. Do any of the confidence intervals include negative values? Why might this be a problem for this type of data?

Question 5

1 points

Create diagnostic plots and comment on:

- normality
- constant variance

Question 6

1 points

Fit a model using a **log transformation** of the response.

Question 7

1 point

Create diagnostic plots for the log-transformed model.

Did things improve?

Question 8

1 point

Compute estimated marginal means for the log model.

Question 9

1 point

Back-transform the means using `exp()`.

Question 10

1 point

Compute the difference between the treatments `Both` and `Control` presented on the log scale then interpret this difference in the data scale

Now we will fit a **GLM appropriate for this type of data** using a **Gamma distribution with a log link**. We will use the model `mod_glm` for the next questions.

```
mod_glm <- glm(clusters ~ treatment,
               family = Gamma(link = "log"),
               data = grape)
```

Question 11

1 point

Create diagnostic plots for the GLM model. Is this model a good fit? Are we concerned about violating assumptions?

Question 12

1 point

Use `emmeans()` to extract treatment means on the **response scale**. Then, comment on the confidence intervals. Do any of the confidence intervals include negative values?

Question 13

1 point

Briefly compare the GLM results to the log-transformed model (are the treatment effects similar?). Discuss how these two approaches are an improvement upon the first approach in which we ignore the non-normality of the residuals.